A Literacy-Free, Report-Free Eye Exam on a Consumer EEG Headband and VR Headset: Toward Objective Vision Screening by Community Health Workers

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***Abstract*—At least a billion people have a vision impairment that could be corrected, and the bottleneck is the screen: an eye test needs a literate, cooperating subject and a trained examiner. We describe an exam that asks nothing of the subject but to look at a dot with one eye covered, on a Muse headband with one electrode added over the visual cortex and a Meta Quest 3S running the stimulus in its browser, under 600 US dollars together. Acuity, contrast, field, colour and the apparent latency come from the steady-state visual evoked potential: an exact test per step, the sweep-VEP threshold rule and the slope of the response phase over flicker rate. A hardware-free phantom with set cut-offs and a dead quadrant passes all twenty-three checks of a full-stack gate. The first sitting on a person fitted no threshold because two electrodes had come loose unnoticed and the stimulus was small; it produced the contact check and the full-window flash reference a device in a health worker's hands must have. Nothing here is a diagnosis.**

***Index Terms*—mobile health; visual screening; SSVEP; sweep VEP; wearable sensors; consumer electronics; community health workers.**

I. The Screen, Not the Lens, Is the Bottleneck

The WHO counts at least 2.2 billion people with a vision impairment, at least a billion of them addressable, uncorrected refractive error first among the causes [1]. A letter chart needs a subject who can read it in the examiner's language and an examiner to judge the answers. The pattern that has carried other screens to community health workers is a consumer device, a worker, and software that checks itself [2], [3]. The steady-state visual evoked potential (SSVEP) gives an objective acuity test with a clinical pedigree, the sweep VEP built for infants, who cannot report [4], and a contrast threshold [5]; it removes literacy, language and examiner bias while bringing biases of its own in hair and headband fit [6].

II. The Rig and the Exam

Mann et al. turned the eye into a camera with the SSVEP [7]; we rebuilt that rig on a Muse headband with a cup electrode at Oz, the phone streaming raw EEG as the hub in the wireless-sensing architecture iCalm set out [8], and a Quest 3S whose browser renders the stimulus (Figs. 1, 2). Every stimulus fills the window: the camera runs had shown the response grows with the flickering area and a small square gives nothing. Each eye opens with the window flickering

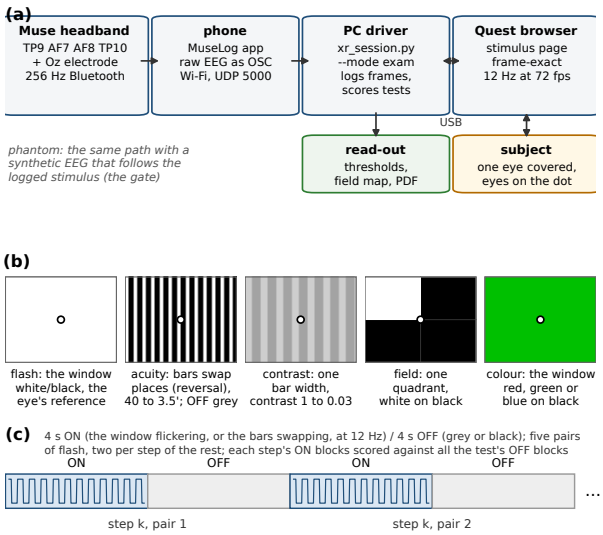


Fig. 1. (a) Scalp to read-out: the headband streams through the phone to the PC, which drives the page in the headset's browser, logs every frame and scores each test; the phantom replaces the first two boxes with a synthetic EEG that follows the logged stimulus. (b) The four stimuli and the OFF screen of the grating tests. (c) One step's blocks.

white/black, the camera's calibration stimulus (five ON/OFF pairs of 4 s at 12 Hz on the 72 Hz display), which must respond and whose channel and amplitude are the reference for the eye's tests. The flash then runs at six rates, 6 to 36 Hz: amplitude against rate is the eye's temporal tuning, the slope of phase against rate its apparent latency [9], whose difference between the eyes is the clinical pattern VEP's measure. Then, two pairs per step: a grating whose bars swap places at the flicker rate, each resolved bar swinging like the flash at a constant mean luminance (pattern reversal [4]), 40 to 3.5 arcmin (acuity) or contrast 1 to 0.03 (contrast); one quadrant at a time (field); the window red, green or blue (colour). Each step's ON blocks are scored against all the test's OFF blocks by the line power at the flicker frequency and its double, with an exact permutation test; a channel above 60 μV rms through a test is set aside. Thresholds follow the sweep-VEP rule: a line through the limb from the step below the finest responding one up the rise, extrapolated to zero. Eight minutes per eye.

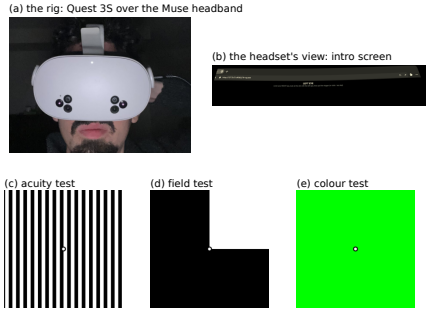


Fig. 2. (a) The Quest 3S over the Muse headband. (b) The headset's own view of one eye's intro screen. (c)–(e) The page during an ON frame of the acuity, field and colour tests.

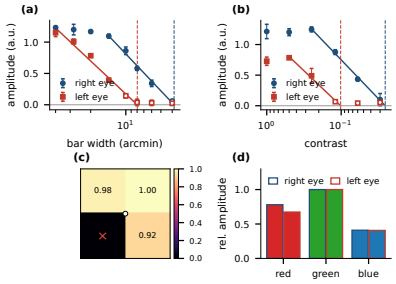


Fig. 3. The gate's read-out. (a) Acuity and (b) contrast per phantom eye (right circles, left squares; open: $p \geq 0.05$), fitted limbs and their zero crossings. (c) The right eye's field, the cross its dead quadrant. (d) Colour planes.

III. Validation without a patient

A device for a worker with no eye clinic behind them must be testable with no patient in front of it. A phantom subject reads the stimulus log as it is written and streams synthetic EEG whose SSVEP follows what is on screen, through a logistic visibility in bar width and contrast with a cut-off per eye, dead quadrants, colour gains, and a latency and a flicker cut per eye. The gate dials in a right eye at 8 arcmin and contrast 0.08 with one dead quadrant, 120 ms late and holding to 40 Hz, a left eye twice as coarse and whole, 150 ms late and half gone by 24 Hz, gains 0.7/1/0.4 for red/green/blue, and asks twenty-three things: every stage delivered, the flash responding in both eyes on Oz and every test read on it, each threshold inside the band where the phantom's visibility runs from 0.02 to 0.6, the left eye at least 1.3× coarser, exactly the dead quadrant named, green $\geq$ red $>$ blue, every rate responding, the half rate and the latencies within 15% and 6 ms of the dialled. All twenty-three pass (Fig. 3): right eye 3.9 arcmin, quadrant 3 silent, a contrast line crossing at 0.026, below the faintest step shown (0.03); left eye 8.0 arcmin and contrast 0.102, half rate 22.3 Hz, its response 29 ms later than the right's.

IV. The first sitting

One of the authors ran the right eye with the left covered, then the left, at 12 Hz. No threshold could be fitted: the only clear response was the full-patch colour flicker (12 Hz line on Oz, $\log_{10}$ SNR $+0.75$ ON against -0.22 OFF); the gratings gave little, the sectors none. Both ear electrodes had sat at 100 to 280 μV rms all sitting, loose under the headset strap, and the session's signal check, written for electrodes at the amplifier rails, had passed them: the quiet failure, not the wrong number, a consumer wearable brings to the field [3]. The exam now measures every channel before each eye, tells the subject which electrode to press in when one is above 60 μV, and names any channel set aside. Its one response was the stimulus closest to the camera's; hence the full window, the flash reference and the reversing bars of Section II.

V. What the same recording carries

A headband records more than the visual response [10]. In companion sittings on the same rig the forehead electrodes gave the blink rate, which stops while a patch flickers, bursts when it goes dark, and doubled between two sittings of one evening, an old marker of fatigue [11]: the hook for a just-in-time adaptive intervention [12], an eye-strain estimate against the subject's own earlier minutes, not detection and not emotion.

VI. Toward the field

Established: the apparatus, an exact test per step, the sweep-VEP threshold rule, the latency from the phase slope, and a phantom gate that catches failures without a volunteer. Not established: any measurement on a person. Open: the headset's pixels per degree, then a blur study with trial lenses against a letter chart, a normative cohort, and a worker-facing flow with the report as one picture.

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